Turn Detection : Whisper Tiny + Classification Head

Approach :- The challenge suggested starting with Whisper Tiny Plus a classification head; use Pretrained Whisper Tiny encoder to extract audio embeddings, then train a classifier on top to Predict whether given 2-second aduio clip ends in a genuine Pause (end of turn) or is mid-utterence (Still speaking, including filler-words Pauses like matlab, toh, acha).

- The encoder itself was kept frozen throughout — no fine-tuning of Whisper's weights. Only a light weight two-layer Classifier (49,538 Parameters) was trained on top of the fixed embeddings. This keeps training fast and avoids the risks of dataset degrading a Pre-trained model's general-Purpose representations.

Data → Used Pipecat-ai/Smart-turn-data-V3.2-train, Dataset linked in the challenge.

270,946 total rows, Labels close to balanced.
Hindi makes up only 4.4% of the dataset (12,006 rows).
Every Hindi row is Synthetic (TTS-generated) — there is no real Hindi speech in this dataset. 82% of the entire dataset is Synthetic.

A Stratified Subset used for training: All Hindi rows, a large sample of rows flagged with mid-Sentence or end-Sentence fillers, and smaller general sample. This keeps the rarer harder cases well represented rather than diluted by the majority class.

Method

1. Load Pretrained Openai/whisper-tiny encoder and froze all its Parameter.
2. For each audio clip, extracted a Pooled 384-dimensional embeddings.
3. Trained a small Classifier (Linear → ReLU → Dropout → Linear) On embeddings, with Class-Weighted Cross-entropy loss to account for label imbalance, and early stopping on validation F1.
4. Ran this twice: Frist on a 1,500-clip subset, then again on full 5,000-clip stratified set, to check whether results held up as the data size increased.

Results:- On 1,520-clip trainning run, evaluated on a held out test split:

Metric:- Accuracy - 0.753, Precision (end-of-turn) - 0.670, Recall (end-of-turn) - 0.913, F1 (end-of-turn) - 0.773 Mid-filler Slice F1 (0.796).

5000-clip trainning run (full Stratified set), same test methodology.

Metric:- Accuracy - 0.800, Precision (end-of-turn) - 0.750, Recall (end-of-turn) - 0.864, F1 (end-of-turn) - 0.803 Mid-filler Slice F1 - 0.748